

Day 17 : Supabase Authentication and Backend

1. Introduction to Supabase

Supabase is an open-source Backend-as-a-Service (BaaS) platform. It provides:

- Authentication
- PostgreSQL Database
- Real-time features
- File Storage
- APIs

It is often called an open-source alternative to Firebase.

2. Why Do We use Supabase?

- No need to build backend from scratch.
- Uses SQL (PostgreSQL).
- Built-in authentication.
- Supports real-time updates.
- Easy integration with Flutter

3. Creating a Supabase Project

Step 1 : Sign Up

Visit:

[Supabase Official Website](https://supabase.com)

Create an account using GitHub or Email.

Step 2 : Create a new project

- Click New Project.
- Select organization.
- Enter:

Project Name

Database Password

Region

- Click Create Project.

4. Get API Keys

After project creation:

Go to:

Settings → API

You will get:

- Project URL
- Anon Public Key

Example

```
url: "https://xyz.supabase.co"  
anonKey: "eyJhbGciOi..."
```

5. Install package

In pubspec.yaml:

```
dependencies:  
  supabase_flutter: ^2.0.0
```

6. Initialize Supabase

```
import 'package:flutter/material.dart';  
import 'package:supabase_flutter/supabase_flutter.dart';  
  
void main() async {  
  WidgetsFlutterBinding.ensureInitialized();  
  
  await Supabase.initialize(  
    url: 'YOUR_PROJECT_URL',  
    anonKey: 'YOUR_ANON_KEY',  
  );  
  
  runApp(MyApp());  
}
```

Access client anywhere:

```
final supabase = Supabase.instance.client;
```

7. Authentication

A. Sign Up

Creates a new user.

```
final response = await supabase.auth.signUp(  
  email: 'abc@gmail.com',  
  password: '123456',  
);
```

B. Sign In

```
final response = await supabase.auth.signInWithPassword(  
  email: 'abc@gmail.com',  
  password: '123456',  
);
```

C. Sign Out

```
await supabase.auth.signOut();
```

D. Get Current User

```
final user = supabase.auth.currentUser;  
  
print(user?.email);
```

E. Magic Link Authentication

User receives login link via email.

```
await supabase.auth.signInWithOtp(  
  email: 'abc@gmail.com',  
);
```

F. OAuth Authentication

Enable Google in:

Authentication → Providers → Google

Code:

```
await supabase.auth.signInWithOAuth(  
  OAuthProvider.google,  
);
```

8 . Using Supabase Database

Supabase uses **PostgreSQL**.

Create a table from Dashboard.

Example table:

students		
id	name	branch
1	Sneha	CS

A. Insert Data

```
await supabase.from('students').insert({
  'name': 'Sneha',
  'branch': 'CS',
});
```

B. Fetch Data

```
final data = await supabase
  .from('students')
  .select();

print(data);
```

C. Fetch with Condition

```
final data = await supabase
  .from('students')
  .select()
  .eq('branch', 'CS');
```

D. Update Data

```
await supabase
  .from('students')
  .update({'branch': 'IT'})
  .eq('id', 1);
```

E. Delete Data

```
await supabase
  .from('students')
  .delete()
  .eq('id', 1);
```

9. Row Level Security (RLS)

RLS controls who can access database rows.

Example:

- User can see only their own data.
- Prevent unauthorized access.

Enable from:

Table Editor → Select Table → Enable RLS

```
CREATE POLICY "Users can view own profile"
ON profiles
FOR SELECT
USING (auth.uid() = id);
```

10. Real-Time Features

Supabase can instantly update UI when database changes.

Examples:

- Chat app
- Notifications
- Live comments

A. Listen For Changes

```
supabase
  .from('messages')
  .stream(primaryKey: ['id'])
  .listen((data) {
    print(data);
  });
```

Example with StreamBuilder

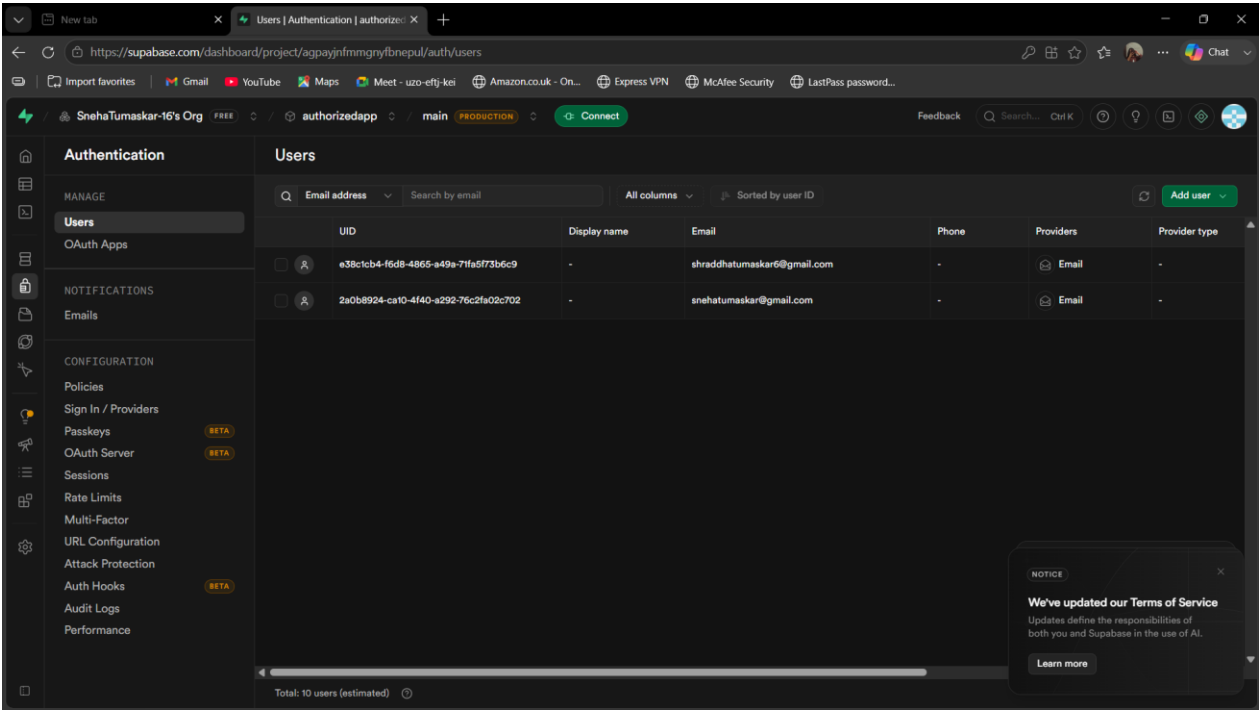
```
StreamBuilder(
  stream: supabase
    .from('messages')
    .stream(primaryKey: ['id']),
  builder: (context, snapshot) {

    if (!snapshot.hasData) {
      return CircularProgressIndicator();
    }

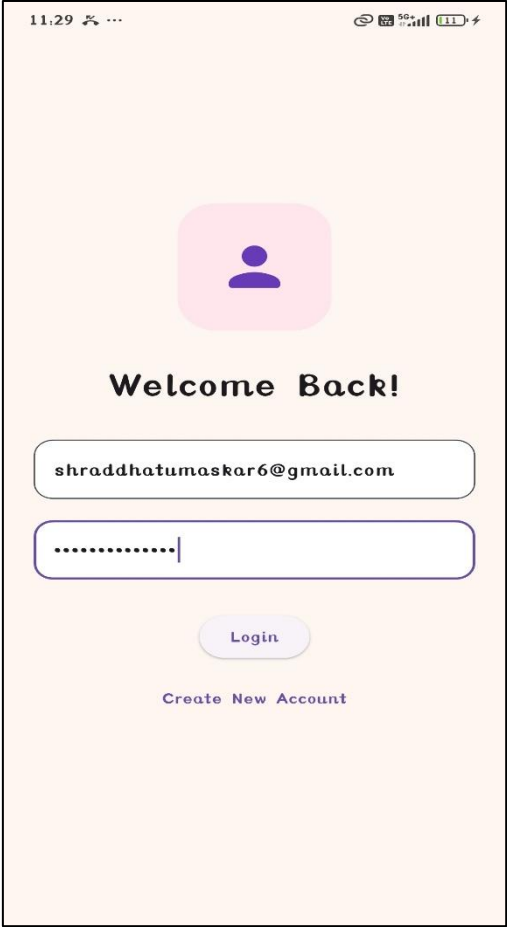
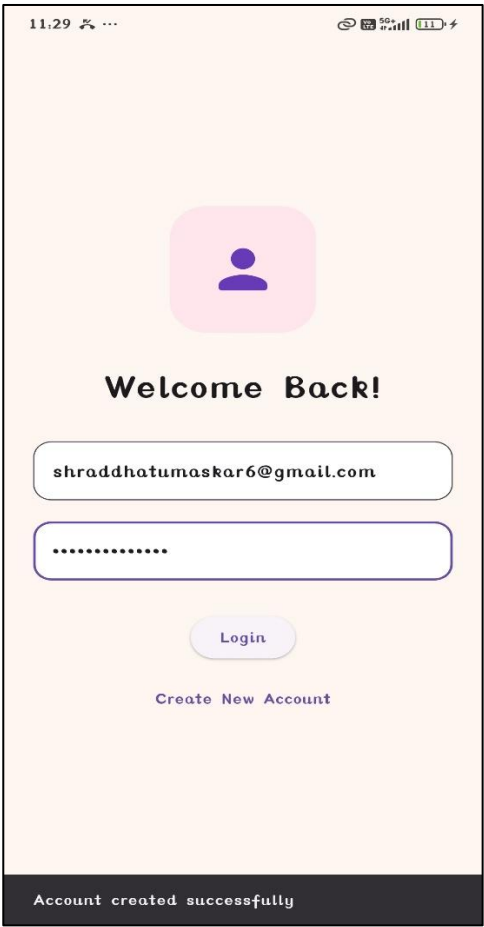
    final messages = snapshot.data!;

    return ListView.builder(
      itemCount: messages.length,
      itemBuilder: (context, index) {
        return Text(messages[index]['text']);
      },
    );
  },
)
```

Authentication of supabase :



Output :



11:29



Profile



shraddhatumaskar6@gmail.com

 Logout